

CHAPTER 7

Fabrication of Semiconductor Devices and ICs

Objectives

You will be able to:

- 1 Describe the process of preparing semiconductor material for device manufacture, and explain diffusion and epitaxial growth.
- 2 Draw sketches to show the fabrication of alloy and diffused diodes.
- 3 Explain the manufacturing requirements for transistors to produce satisfactory current gain, frequency response, power dissipation, etc.
- 4 Describe the various transistor fabrication methods and their performance characteristics.
- 5 Discuss manufacturing methods for the production of integrated circuits.
- 6 Draw sketches to show the construction of monolithic IC components, and discuss problems that occur with the use of IC components.
- 7 Sketch various diode, transistor, and IC packages, and show the terminal identification systems.

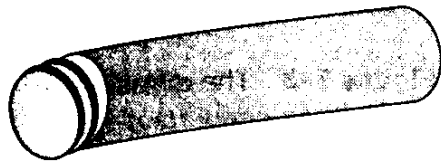
INTRODUCTION

The method employed to manufacture a semiconductor device determines its electrical characteristics and, thus, dictates its applications. For example, low-current fast-switching diodes and transistors must be fabricated differently from high-power devices. The process that produces a transistor with high current gain may result in undesirable large junction capacitances. A technique that gives a small junction capacitance may produce devices with low breakdown voltages. An *integrated circuit (IC)* consists of many transistors, diodes, and other components in one package. Usually, all the components are fabricated on one small silicon chip.

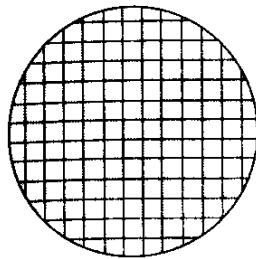
7-1 PROCESSING OF SEMICONDUCTOR MATERIALS

Preparation of Silicon and Germanium

Silicon is one of the commonest elements on earth. It occurs as silicon dioxide (SiO_2) and as silicates, compounds of silicon and other materials. Germanium



(a) Semiconductor bar sliced into wafers



(b) Scribed wafer

Figure 7-1 Bars of single-crystal semiconductor material are sliced into wafers for the manufacture of semiconductor devices.

is derived from zinc or copper ores. When converted to bulk metal, silicon and germanium contain large quantities of impurities. Both metals must be carefully refined before they can be used for the manufacture of semiconductor devices.

Semiconductor material is normally *polycrystalline* after it is refined. This means that it is made up of many individual formations of atoms with no overall fixed pattern. For use in semiconductors, the metal must be converted into *single-crystal* material; that is, all of its atoms must be arranged into a single pattern.

In its final condition for device manufacture, silicon and germanium are in the form of single-crystal bars about 2.5 cm in diameter and perhaps 30 cm long. The bars are sliced into disc-shaped wafers about 0.4 cm thick, and the wafers are polished to a mirror surface.

Several thousand devices are usually fabricated on the surface of each wafer; then the wafers are scribed and cut like glass (see Fig. 7-1).

Diffusion

For processing, semiconductor wafers contained in an enclosure are heated by means of radio frequency (RF) heating coils. This is illustrated in Fig. 7-2. When

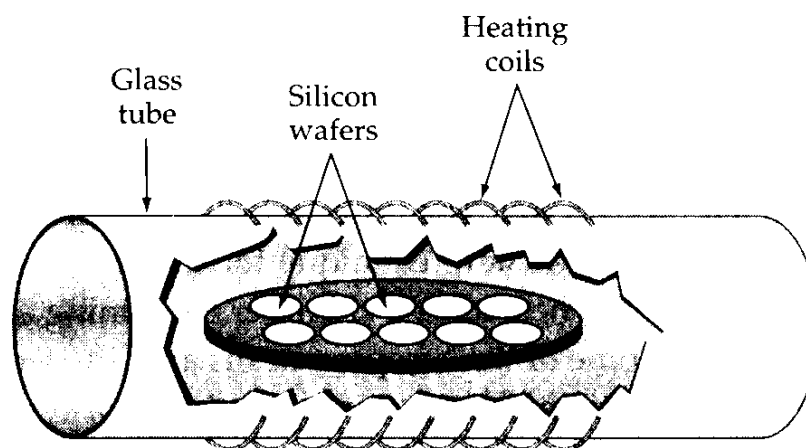


Figure 7-2 Semiconductor wafers are heated in an atmosphere containing *n*-type or *p*-type impurities that diffuse into the surface of the wafers.

wafers of n -type material are raised to a high temperature in an atmosphere containing p -type impurity atoms, some of the impurities *diffuse* into each wafer. This converts the outer layer of the n -type material into p -type (Fig. 7-3). The diffusion process can be continued by further heating the wafers in an atmosphere containing n -type impurity atoms, so that npn layers are produced, as illustrated. Because the diffusion process is very slow (about $2.5\ \mu\text{m}$ per hour), very narrow diffused regions can be created by careful timing.

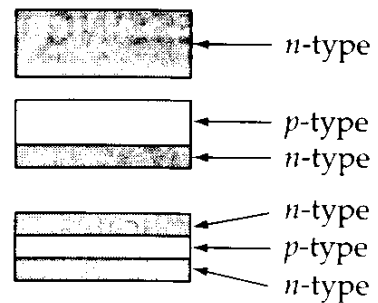


Figure 7-3 The diffusion process converts n -type semiconductor into p -type, and vice versa.

Epitaxial Growth

The *epitaxial* process is very similar to the diffusion process, except that silicon or germanium atoms are contained in the gas surrounding the semiconductor wafers. The semiconductor atoms in the gas grow (accumulate) on the wafer in the form of a thin layer (Fig. 7-4). This layer is single-crystal material and it may be p -type or n -type, according to the impurity content of the gas. The epitaxial layer may be 'doped' by the diffusion process.

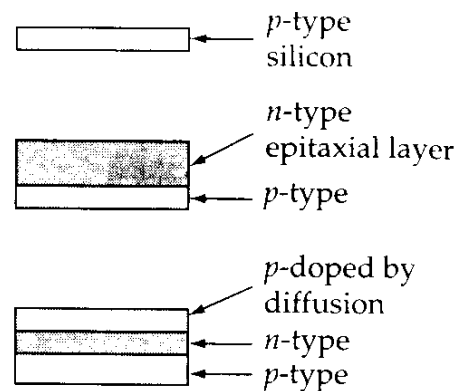


Figure 7-4 Epitaxial growth adds layers of n -type or p -type material to the surface of a wafer.

Section 7-1 Review

7-1.1 Briefly explain polycrystalline material, single-crystal material, diffusion, and epitaxial growth.

7-2 DIODE FABRICATION AND PACKAGING

Alloy and Diffused Diodes

Two commonly used techniques for diode manufacture are the *alloy* method and the *diffusion* method. To construct an alloy diode, a pn -junction is formed by melting a tiny pellet of aluminum (or some other p -type impurity) on the surface of an n -type crystal. Alternatively, an n -type impurity may be melted on the surface of a p -type crystal. The process is illustrated in Fig. 7-5a.

Figure 7-5b shows the diffusion technique for diode construction. When an n -type semiconductor is heated in a chamber containing an acceptor impurity in vapour form, some of the acceptor atoms are diffused (or absorbed) into the n -type crystal. This produces a p -region in the n -type material and thus creates a pn -junction. The size of the p -region can be precisely defined by uncovering only part of the n -type material during the diffusion process (the

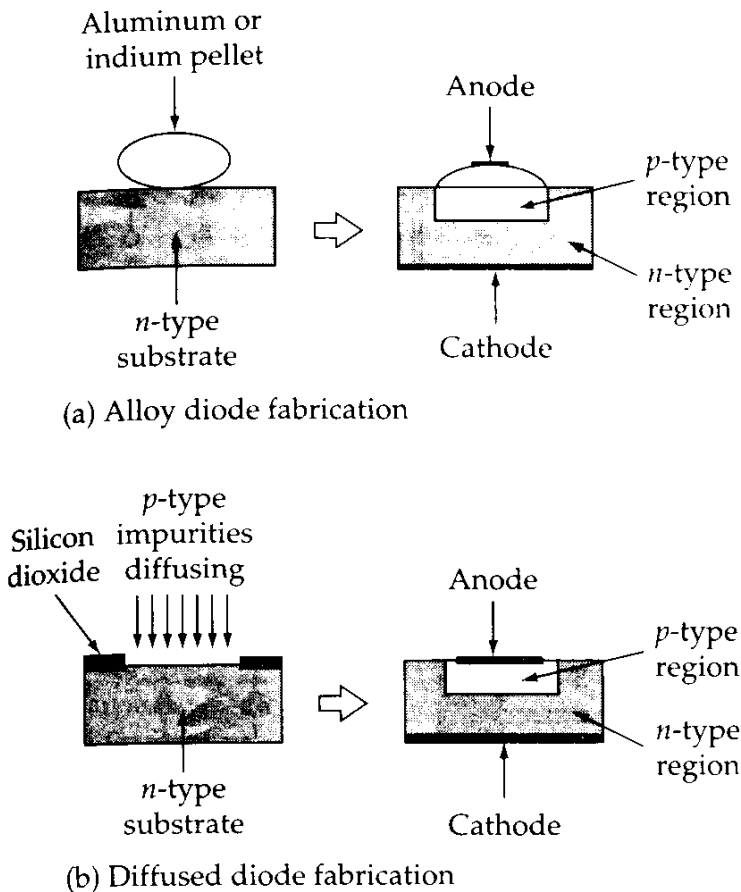


Figure 7-5 Diode fabrication methods. For the alloy diode, a pellet containing p -type impurities is melted into n -type material. A diffused diode is created by heating a wafer of n -type material in an atmosphere containing p -type impurities.

rest of the surface has a thin coating of silicon dioxide). Metal contacts are electroplated onto each region for connecting leads.

The diffusion technique lends itself to the simultaneous fabrication of many hundreds of diodes on one small disc of semiconductor material. This process is also used in the production of transistors and integrated circuits.

Diode Packaging

As explained in Section 2-1, semiconductor diodes vary widely in size, depending on their application. Figure 7-6 shows typical single diode packages for low-, medium-, and high-current devices. In Fig. 7-7 multiple diode packages are illustrated. Two diodes in a surface mount package (see Section 7-7) are shown in Fig. 7-7a, and two in a power-transistor-type package are illustrated in Fig. 7-7b. Note from the circuits that the cathode terminals are internally commoned in each case.

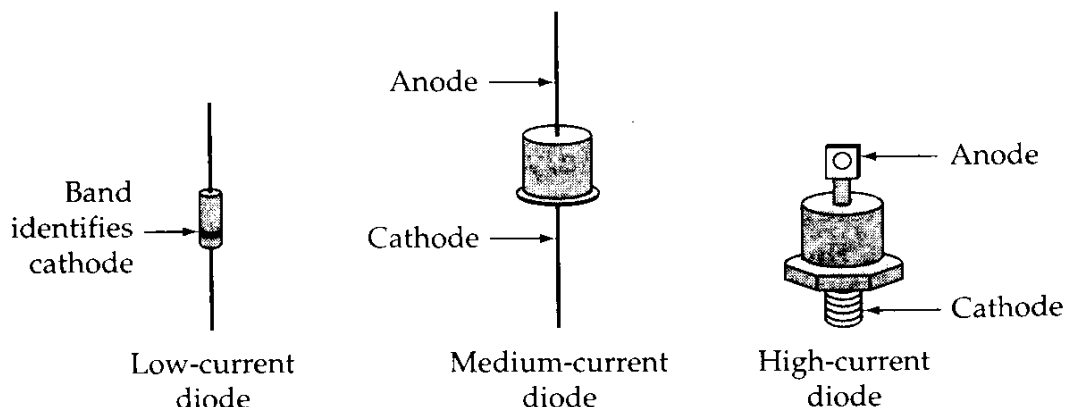


Figure 7-6 Individual diodes vary in size according to the current they are required to pass.

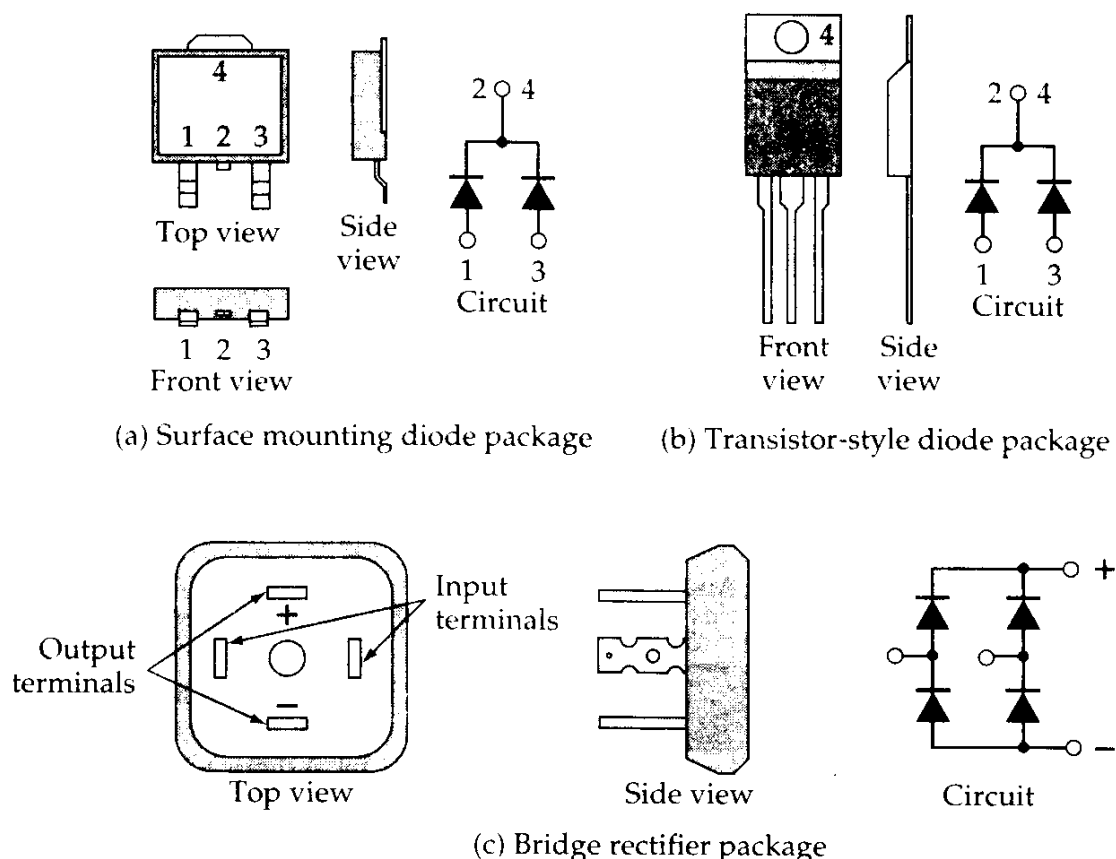


Figure 7-7 Some multiple diode packages.

A rectifier assembly is shown in Fig. 7-7c. This consists of four diodes connected to function as a bridge rectifier and contained in one package. One side of the package has a metal plate for mounting on a heat sink (see Section 8-8). The two output terminals are identified as + and -, and the other two terminals are for the ac input.

Section 7-2 Review

7-2.1 Briefly describe the construction of alloy diodes and diffused diodes.

7-3 TRANSISTOR CONSTRUCTION AND PERFORMANCE

Current Gain

Good current gain requires that most charge carriers from the emitter pass rapidly to the collector. Thus there should be little outflow of charge carriers through the base terminal and few carrier recombinations within the base. These conditions dictate a very narrow, lightly doped base (see Fig. 7-8).

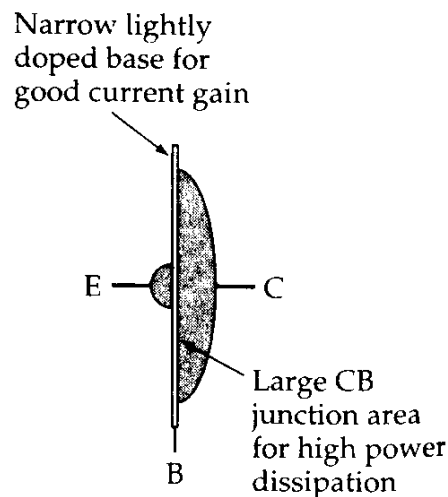


Figure 7-8 High current gain requires a narrow lightly doped transistor base, and high power dissipation demands a large CB junction area.

High Power

High-power transistors must have large emitter-base surfaces to provide the required quantity of charge carriers. Large collector-base surfaces are also needed to dissipate power without overheating the collector-base junctions (Fig. 7-8).

Frequency Response

For the highest possible frequency response, the transistor base region must be very narrow to ensure a short transit time of charge carriers from emitter to collector (see Fig. 7-9). Input capacitance must also be kept to a minimum, and this requires a small-area emitter-base junction as well as a highly resistive (lightly doped) base region.

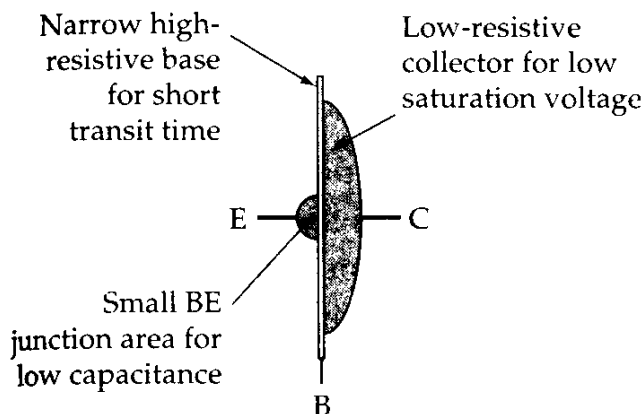


Figure 7-9 A small BE junction area and narrow base are required for high frequency performance and fast switching. Low saturation voltage is promoted by a low-resistive collector region.

Because power transistors require large junction surfaces, and high-frequency performance demands small junction areas, there is a conflict in high-frequency power transistors. To keep the junction area to a minimum and still provide adequate charge-carrier emission, the emitter-base junction is usually in the form of a long thin zigzag strip.

Switching Transistors

Fast switching demands the same low junction capacitance required for good high-frequency performance. A switching transistor should have a low *saturation voltage* and a short storage time (see Section 8-5). For low saturation voltages, the collector region must have low resistivity (Fig. 7-9). Short storage time requires fast recombination of charge carriers left in the depletion region at the collector-base junction. Fast recombination is helped by additional doping of the collector with gold atoms.

Breakdown and Punch-Through

The transistor collector-base junction is normally reverse biased, so the maximum collector voltage is limited by the junction reverse breakdown voltage. To achieve high breakdown voltages, either the collector or the base must be very lightly doped. If the base is more lightly doped than the collector, the collector-base depletion region penetrates deeply into the base. This will cause transistor breakdown by *punch-through* when it links with the emitter-

base depletion region. To avoid punch-through, the collector region close to the base is usually more lightly doped than the base. The collector-base depletion region then spreads into the collector rather than into the base.

Section 7-3 Review

- 7-3.1 Briefly discuss the requirements for constructing transistors with (a) good current gain and (b) high-frequency response.
- 7-3.2 Explain the design requirements for transistors with (a) high power dissipation and (b) high breakdown voltage.
- 7-3.3 List the requirements for switching transistor construction.

7-4 TRANSISTOR FABRICATION

Alloy Transistors

For manufacture of *alloy transistors*, single-crystal *n*-type wafers are scribed into many small sections, or *dice*, each of which forms the substrate for one transistor. A small pellet of *p*-type material is melted on one surface of each section until it partially penetrates and forms an alloy with the substrate (Fig. 7-10a), thus creating a *pn*-junction. The process is repeated on the other side of the wafer to constitute a *pnp* transistor.

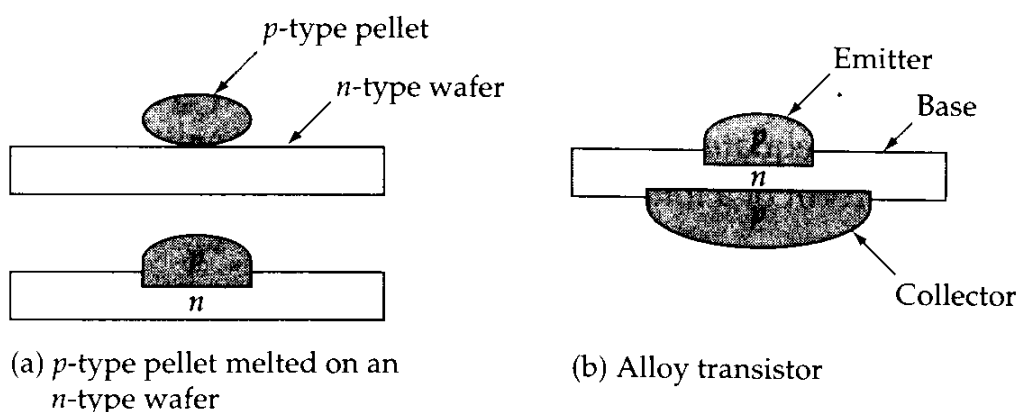


Figure 7-10 A *pnp* alloy transistor is constructed by melting *p*-type pellets on the surface of an *n*-type wafer.

One of the junctions has a large area and one has a small area, as illustrated in Fig. 7-10b. The small-area junction becomes the emitter-base junction, and the larger one becomes the collector-base junction. One reason for this is that the large-area junction most easily collects all the charge carriers emitted from the small-area junction. Another more important reason is that most of the power is dissipated at the transistor collector-base junction.

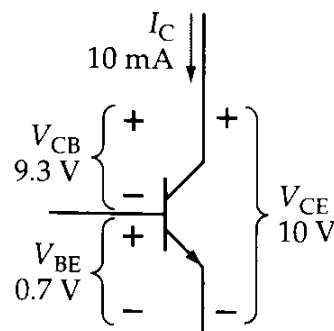


Figure 7-11 Most of the power dissipation in a transistor occurs at the collector-base junction.

Suppose that a silicon transistor has $I_C = 10 \text{ mA}$ and $V_{CE} = 10 \text{ V}$ (see Fig. 7-11). The total power dissipated in the transistor is

$$\begin{aligned} P &= V_{CE} \times I_C = 10 \text{ V} \times 10 \text{ mA} \\ &= 100 \text{ mW} \end{aligned}$$

The base-emitter voltage $V_{BE} = 0.7 \text{ V}$, and the collector-base voltage is

$$\begin{aligned} V_{CB} &= V_{CE} - V_{BE} = 10 \text{ V} - 0.7 \text{ V} \\ &= 9.3 \text{ V} \end{aligned}$$

The power dissipated at the base-emitter junction is

$$\begin{aligned} V_{BE} \times I_C &= 0.7 \text{ V} \times 10 \text{ mA} \\ &= 7 \text{ mW} \end{aligned}$$

and the power dissipated at the collector-base junction is

$$\begin{aligned} V_{CB} \times I_C &= 9.3 \text{ V} \times 10 \text{ mA} \\ &= 93 \text{ mW} \end{aligned}$$

Microalloy Transistors

Because very narrow base widths are difficult to obtain with alloy transistors, they are not suitable for high-frequency applications. To improve the high-frequency performance, holes are first etched into the wafer from each side, leaving a very thin layer (Fig. 7-12). By a plating process, surfaces of impurity material are formed on each side of the thin n -type portion. Heat is then applied to alloy the impurities into the base region. This process produces very thin base regions and good high-frequency performance. The device is termed a *microalloy* transistor.

Microalloy-Diffused Transistors

In microalloy transistors, the collector-base depletion region penetrates deeply into the very thin base. Thus, a major disadvantage is that punch-through can occur at very low collector voltages.

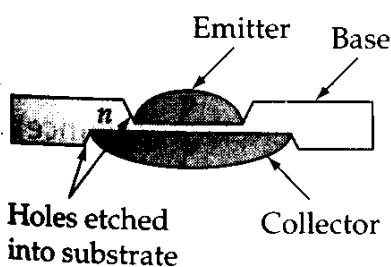


Figure 7-12 A very narrow base region gives microalloy transistors good high-frequency performance.

Microalloy-diffused transistors use a substrate that is initially undoped. After the holes are etched into each side of the wafer to produce the thin base region, the base is doped by diffusion from the collector side. The diffusion can be carefully controlled so that the base region is heavily doped at the collector side, with the doping becoming progressively less until the material is almost intrinsic (un-

doped) at the emitter. With this kind of doping, the collector-base depletion region penetrates only a short distance into the base, so that high punch-through voltages are achieved.

Diffused Mesa

In the production of *mesa transistors*, several thousand transistors are simultaneously formed on the wafer by the diffusion process. As shown in Fig. 7-13, the main body of the wafer becomes the *n*-type collectors, the diffused *p*-regions become the bases, and the final *n*-regions are the emitters. Metal films are deposited on the base and emitter surfaces for contacts.

The transistors could be separated by the usual process of scribing lines on the surface of the wafer and breaking it into individual units. However, this produces very rough edges that result in relatively high leakage currents between collector and base. So, before cutting the wafer, the transistors are isolated by etching away the unwanted portions of the diffused area to form troughs between devices. As illustrated, the base and emitter regions now project above the main wafer, which forms the collector region. This is the mesa structure. The narrow base widths that can be achieved by the diffusion process make the mesa transistor useful at very high frequencies.

Epitaxial Mesa

One of the disadvantages of the process just described is that, because the collector region is highly resistive, diffused mesa transistors have a high saturation voltage (see Section 4-4). Such devices are unsuitable for saturated switching applications. This same characteristic (high collector resistance) is desirable for high punch-through voltage. One method of achieving both high punch-through voltage and low saturation level uses the epitaxial process to produce an *epitaxial mesa transistor*.

Starting with a low-resistive (highly doped) wafer, a thin, highly resistive epitaxial layer is grown. This layer becomes the collector, and the base and emitter are diffused as already discussed. The arrangement is illustrated in

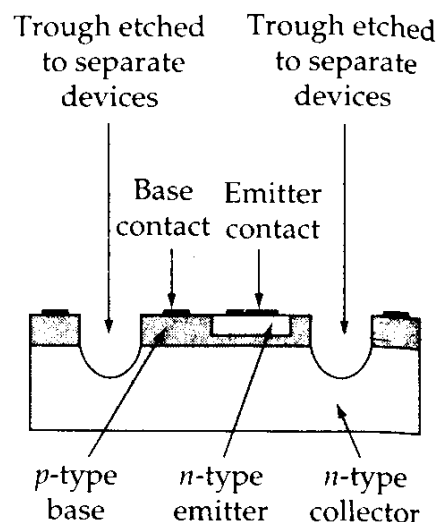


Figure 7-13 Mesa transistors are created by etching troughs between individual devices to give low collector-base leakage currents.

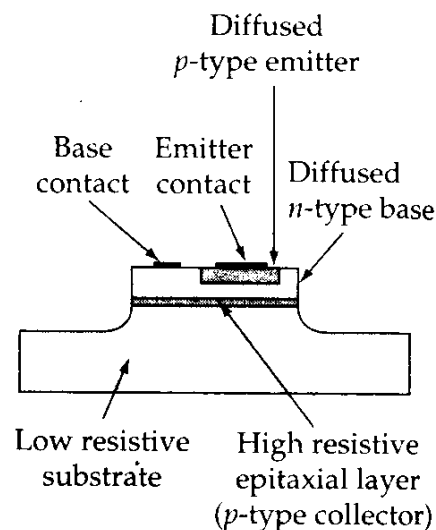


Figure 7-14 Epitaxial mesa transistor. The epitaxial process, combined with the mesa structure, gives high breakdown voltages and low saturation levels.

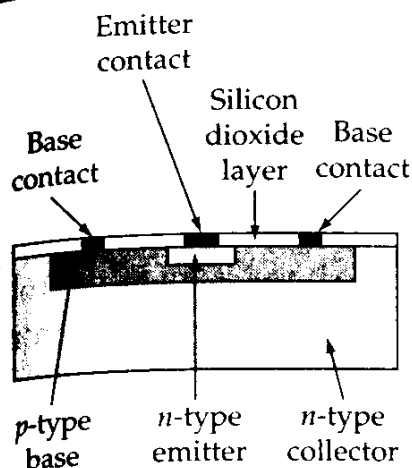


Figure 7-15 A diffused planar transistor has the collector-base junction covered, so that I_{CBO} is reduced to very low levels.

Fig. 7-14. The punch-through voltage is high because the collector-base depletion region spreads deepest into the lightly doped collector. Saturation voltage is low because the collector region is very narrow and the main body of the wafer through which collector current must flow has a very low resistance.

Diffused Planar Transistor

In all the transistors described previously, the collector-base junction is exposed (within the transistor package), and substantial charge carrier leakage can occur at

the junction surface. In the *diffused planar* transistor illustrated in Fig. 7-15, the collector-base junction is covered with a layer of silicon dioxide. This construction gives a very low collector-base leakage current (I_{CBO}), typically 0.1 nA.

Annular Transistor

A problem that occurs particularly with *pnp* planar transistors is the *induced channel*. This results when a relatively high voltage is applied to the silicon dioxide surface (at one of the terminals). Consider the *pnp* structure shown in Fig. 7-16a. If the surface of the silicon dioxide has a positive voltage, minority charge carriers within the lightly doped *p*-type substrate are attracted by the positive potential. The minority charge carriers concentrate at the upper edge of the substrate and form an *n*-type channel from the base to the edge of

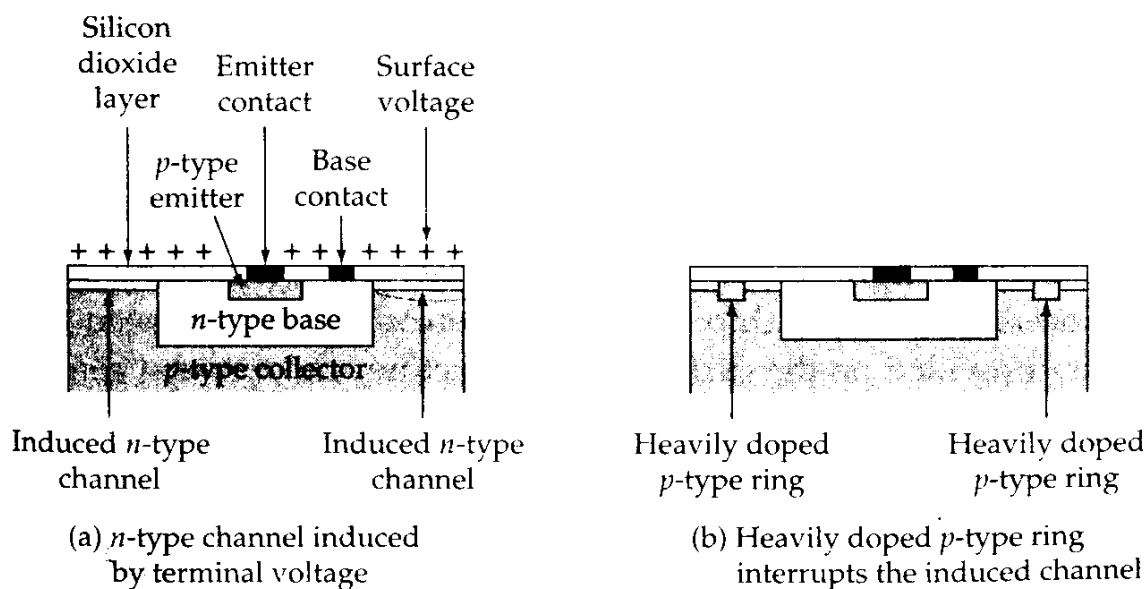


Figure 7-16 Annular transistors have a heavily doped *p*-type ring included to interrupt unwanted *n*-type channels that are induced by high voltages.

the device. This becomes an extension of the n -type base region and results in charge carrier leakage at the exposed edge of the collector-base junction.

The problem arises because the p -type substrate is highly resistive. If it were heavily doped with p -type charge carriers, the concentration of n -type carriers would be absorbed (the electrons would be swallowed by holes). The introduction of a heavily doped p -type ring around the base, as in Fig. 7-16b, interrupts the induced channel and isolates the collector-base junction from the device surface. The *annular transistor*, therefore, is a high-voltage device with the low collector-base leakage of the planar transistor.

Section 7-4 Review

- 7-4.1 Briefly explain the construction of alloy, microalloy, and microalloy-diffused transistors.
- 7-4.2 Describe diffused mesa and epitaxial mesa transistor construction.
- 7-4.3 Discuss the construction of diffused planar and annular transistors.

7-5 INTEGRATED CIRCUITS

IC Types

An *integrated circuit* (IC) consists of several interconnected transistors, resistors, and so on, all contained in one small package with external connecting terminals. The circuit may be entirely self-contained, requiring only input and output connections and a supply voltage. Alternatively, a few external components may have to be connected to make the circuit operate.

Integrated circuits may be classified in terms of their function as *analog* or *digital*. Analog ICs (also termed *linear* ICs) may be amplifiers, voltage regulators, and so on. Digital ICs (logic gates and counting circuits) contain transistors that are normally either in a switched-off or switched-on state.

The techniques used in manufacturing integrated circuits provide another method of classification. The major IC manufacturing technique is termed *monolithic*. Other fabrication techniques are *thin-film*, *thick-film*, and *hybrid*.

Monolithic Integrated Circuits

In a *monolithic* integrated circuit all components are fabricated by the diffusion process on a single chip of silicon. Interconnections between components are provided on the surface of the structure, and external connecting wires are taken out to terminals, as illustrated in Fig. 7-17. The vast majority of integrated circuits are monolithic because the diffusion process is the most economical for mass production.

Thin Film

Thin-film integrated circuits are constructed by depositing films of conducting material on the surface of a glass or ceramic base. By controlling the

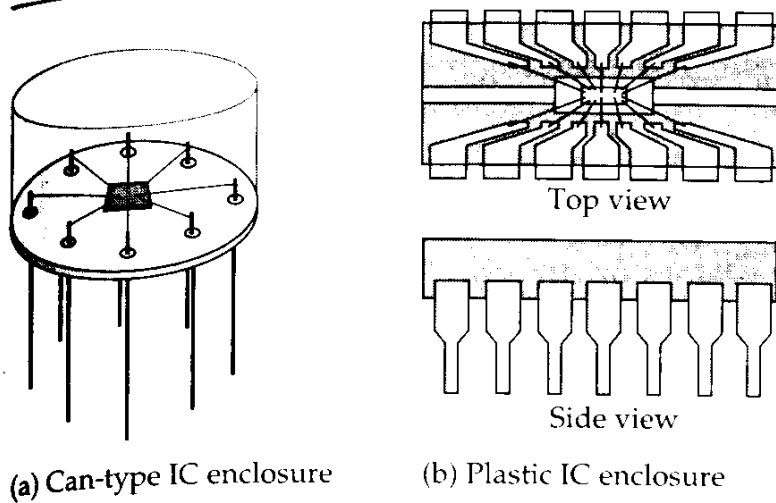


Figure 7-17 Monolithic integrated circuits consist of a single-chip circuit contained in a metal or plastic enclosure with connecting pins.

width and thickness of the films, and by using different materials selected for their resistivity, resistors and conductors are fabricated. Capacitors are produced by sandwiching a film of insulating oxide between two conducting films. Inductors are made by depositing metal film in a spiral formation. Transistors and diodes are usually tiny discrete components connected into the circuit.

Thick Film

Thick-film integrated circuits are sometimes referred to as *printed thin-film circuits*. In this process, silk-screen printing techniques are employed to create the desired circuit pattern on a ceramic substrate. The inks used are pastes which have conductive, resistive, or dielectric properties. After printing, the circuits are fired at a high temperature in a furnace to fuse the films to the substrate. Thick-film passive components are fabricated in the same way as those in thin-film circuits. As with thin-film circuits, active components are added as separate devices. A portion of a thick-film circuit is shown in Fig. 7-18.

Integrated circuits produced by thin- or thick-film techniques usually have better component tolerances and give better high-frequency performance than monolithic integrated circuits.

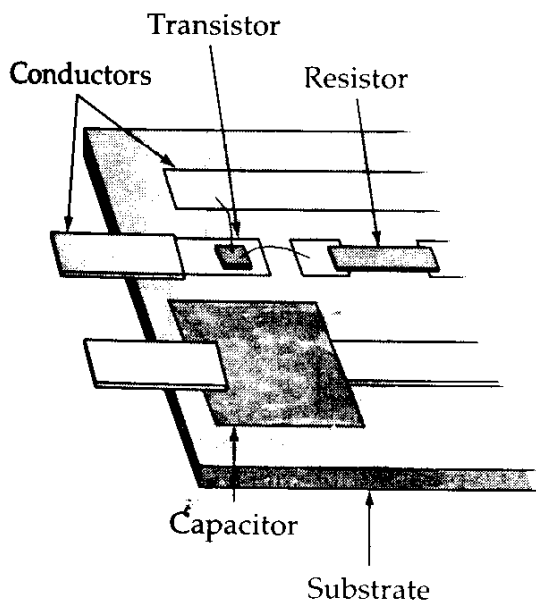


Figure 7-18 Enlarged portion of a thick film integrated circuit.

Hybrid

Figure 7-19 illustrates the structure of a *hybrid* or *multichip* integrated circuit. As the name implies, the circuit is constructed by interconnecting a number of individual chips. The active components are diffused transistors or diodes. The passive components may be groups of diffused resistors or capacitors on a single chip, or they may be thin-film components. Connections between chips are provided by fine wire or metal film.

Like thin-film and thick-film ICs, multichip circuits usually have better performance than monolithic circuits. Although the process is too expensive for mass production, multichip techniques are quite economical for small quantities and are frequently used as prototypes for monolithic integrated circuits.

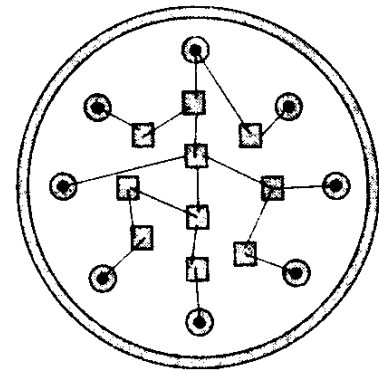


Figure 7-19 Uncovered top view of hybrid integrated circuit. Individual components are interconnected to form a single circuit.

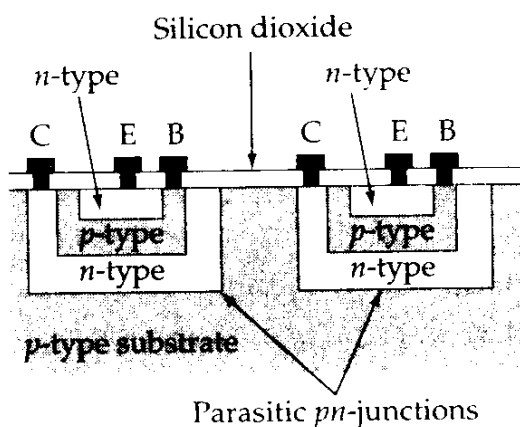
Section 7-5 Review

7-5.1 Briefly define monolithic, thin-film, thick-film, and hybrid integrated circuits.

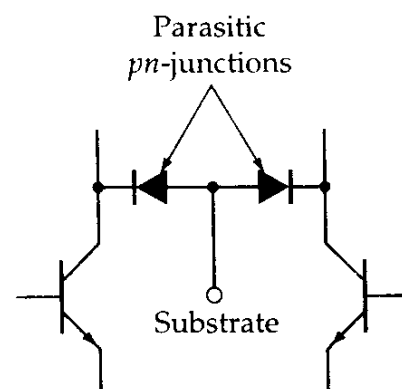
7-6 IC COMPONENTS AND CIRCUITS

Transistors and Diodes

The epitaxial planar diffusion process described in Section 7-4 is normally employed for the manufacture of IC transistors and diodes. Collector, base, and emitter regions are diffused into a silicon substrate, and surface terminals are provided for connection, as illustrated in Fig. 7-20a.



(a) Cross-section of two adjacent IC transistors



(b) Circuit diagram showing parasitic junctions

Figure 7-20 Integrated circuit components have unwanted (parasitic) interconnecting *pn*-junctions that must be kept reverse biased.

In discrete transistors the substrate is normally used as a collector. If this were done with transistors in a monolithic integrated circuit, all transistors fabricated on one substrate would have their collectors connected together. For this reason, separate collector regions must be diffused into the substrate.

Even though separate collector regions are formed, they are not completely isolated from the substrate. Figure 7-20a shows that an unwanted (*parasitic*) pn -junction is formed by the substrate and the transistor collector region. If the circuit is to function correctly, these junctions must never become forward-biased. Thus, in the case of a p -type substrate, the substrate must always be kept negative with respect to the transistor collectors (Fig. 7-20b). This requires that the substrate be connected to the most negative terminal of the circuit supply.

The parasitic junctions can affect the circuit performance even when they are reverse biased. The junction reverse leakage current can be a serious problem in circuits that operate at very low current levels. The capacitance of the reverse biased junction can affect the circuit high-frequency performance, and the junction breakdown voltage imposes limits on the usable level of supply voltage. All these factors can be minimized by using highly resistive (lightly doped) material for the substrate.

Integrated circuit diodes are fabricated by diffusion exactly like transistors. Two of the regions are used to form one pn -junction. Alternatively, the collector region of a transistor may be connected directly to the base (Fig. 7-21), so that the device behaves as a diode while operating similarly to a saturated transistor (see Section 4-4).

Circuits

Some IC internal circuitry differs substantially from discrete component circuits. This is because of similarities between devices that are all fabricated at the same time on the same substrate. One such circuit, known as a *current mirror* is illustrated in Fig. 7-22. As explained above, BJTs are frequently used as diodes in integrated circuits. The current mirror uses such a diode to con-

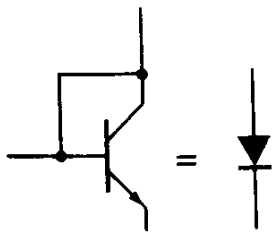


Figure 7-21 A transistor can be made to behave as a diode by connecting the collector and base.

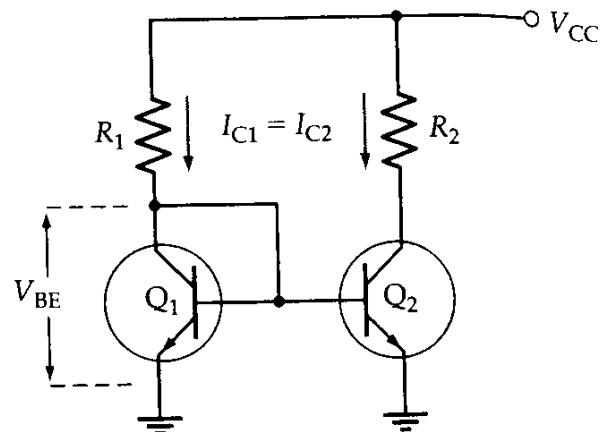


Figure 7-22 Current mirror. Q_1 is diode-connected, and because $V_{BE2} = V_{BE1}$, $I_{C2} = I_{C1}$.

trol the current level in an adjacent BJT. Transistor Q_1 in Fig. 7-22 is connected to function as a diode, with its base and emitter terminals commoned with Q_2 , which functions as a transistor. The voltage drop across Q_1 (collector and base to emitter) is identified as V_{BE} , and so the current through Q_1 is

$$I_{C1} = \frac{V_{CC} - V_{BE}}{R_1}$$

V_{BE} is usually 0.7 V, but the precise value of V_{BE1} depends upon the level of I_{C1} . Because the base and emitter terminals of Q_1 and Q_2 are commoned, V_{BE1} and V_{BE2} are the same voltage. Also, because Q_1 and Q_2 are identical devices, each transistor will have the same level of collector current when they have the same base-emitter voltage. Consequently, I_{C2} (the collector current in Q_2) *mirrors* the current in Q_1 . If I_{C1} is altered by adjustment of R_1 , I_{C2} changes to match the change in I_{C1} . In this situation Q_2 is biased to any desired level of I_C without the use of a voltage divider or an emitter resistor.

Resistors

The resistivity of semiconductor material is a function of doping density, and so resistors can be produced by doping strips of material and providing terminals (Fig. 7-23). The range of resistor values that may be produced by the diffusion process varies from ohms to hundreds of kilohms. The typical tolerance, however, may be no better than $\pm 5\%$ and may even be as high as $\pm 20\%$. However, if all resistors are diffused at the same time, then the *tolerance ratio* can be good. For example, several resistors with the same nominal value may all have a $\pm 20\%$ tolerance but have actual resistance values within a few percent of each other.

Another method of producing resistors for integrated circuits uses the thin-film technique. In this process a metal film is deposited on a glass or silicon dioxide surface. The thickness, width, and length of the film are regulated to give a desired resistance value. Since diffused resistors can be processed while diffusing transistors, the diffusion technique is the least expensive and the most frequently used.

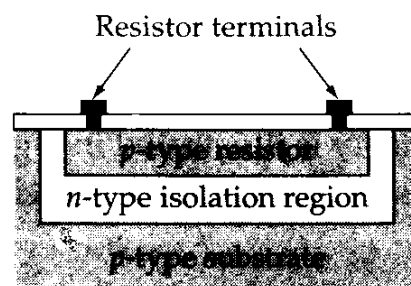


Figure 7-23 A monolithic IC resistor is created by doping a strip of semiconductor material to give the desired resistance.

Capacitors

Because all pn -junctions have capacitance, capacitors may be produced by fabricating suitable junctions. As in the case of other diffused components, parasitic junctions are unavoidable. Both the parasitic and the main junction must be kept reverse biased to avoid direct current flow. The depletion

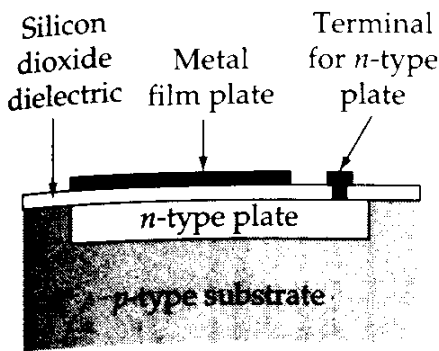


Figure 7-24 IC capacitor using silicon dioxide as the dielectric. The lower plate is a heavily doped n -type semiconductor, and the upper plate is a metal film.

region width and junction capacitance vary with changes in reverse bias; so for capacitance value stability, a dc reverse bias greater than maximum signal voltage must be maintained across the junction.

Integrated circuit capacitors may also be fabricated by using the silicon dioxide surface layer as a dielectric. A heavily doped n -region is diffused to form one plate of the capacitor. The other plate is created by depositing a film of aluminum on the silicon dioxide (see Fig. 7-24). Voltages of any polarity may be employed with this type of capacitor, and the breakdown

voltage is very much larger than that for diffused capacitors. The junction areas available for creation of IC capacitors are very small indeed, so that normally only picofarad capacitance values are possible.

Section 7-6 Review

7-6.1 Discuss the fabrication of integrated circuit transistors and diodes.

7-6.2 Explain how IC resistors and capacitors are constructed.

7-7 TRANSISTOR AND IC PACKAGING

Discrete Transistor Packaging

Many low-power transistors are encapsulated in resin with protruding metal connecting leads, as illustrated in Fig. 7-25. This is known as a TO-92 package. Note the emitter, base, and collector terminal connections. These are in the sequence E, B, and C, from left to right, if we look at the bottom of the transistor with the flat side uppermost. This kind of package is economical, but it has a limited operating temperature range.

Figure 7-26 shows another method of low-power transistor packaging where the device is hermetically sealed in a metal can. Depending on the size of the can, this is identified as a TO-5 to TO-18 package. The transistor is first mounted with its collector in mechanical and electrical contact with a heat-conducting metal plate. Wires (insulated from the plate) pass through for the emitter and base connections, and the covering metal can is welded to

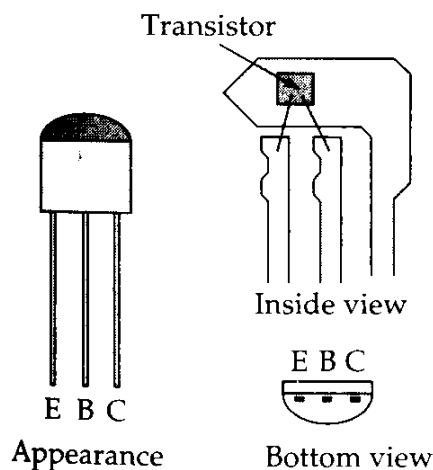


Figure 7-25 Low-power resin-encapsulated transistor; TO-92 package.

the plate. Looking at the bottom of the transistor, and moving clockwise from the tab, the terminals are identified as E, B, and C. The metal can enclosure gives a greater temperature range and greater power dissipation than resin encapsulation.

For high-power transistor packaging, a sealed can (TO-3) is often used (Fig. 7-27a). In this case the heat-conducting plate is much larger than in the TO-5 package, and it is designed for mounting directly on a heat sink. Connecting pins are provided for the base and the emitter, and the collector connection

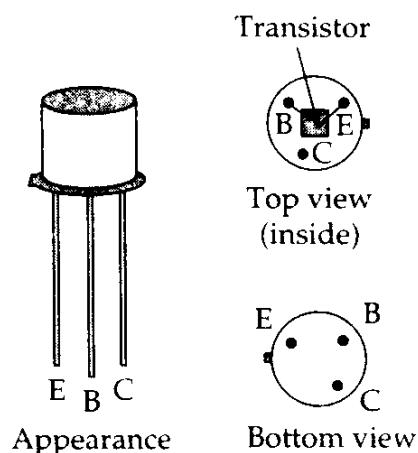
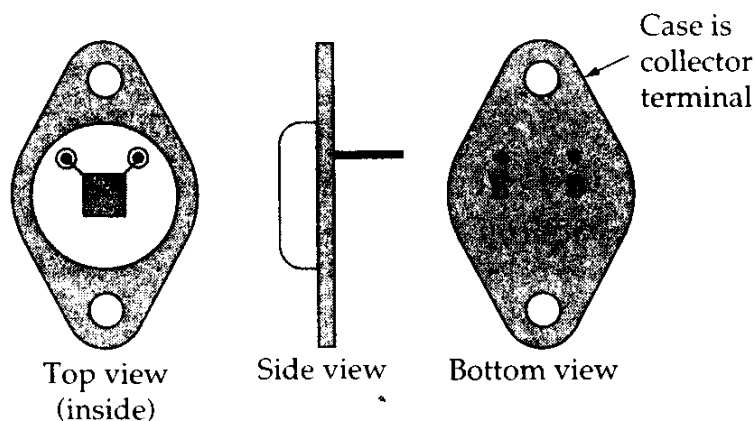
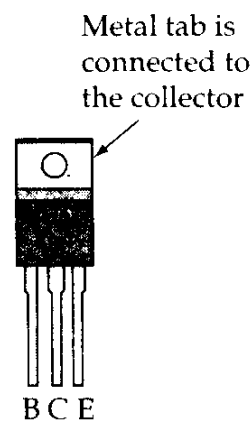


Figure 7-26 Low-power transistor, enclosed in a metal can; TO-18 package.



(a) TO-3 metal can enclosure



(b) TO-220 plastic enclosure

Figure 7-27 Metal can-type and plastic-encapsulated high-power transistors.

is made by means of the metal plate. Note the terminal identifications, again looking at the bottom of the device.

Figure 7-27b shows a plastic-encapsulated power transistor with a metal tab for fastening to a heat sink. Once again, note the transistor terminal identifications, and note that the the metal tab is connected to the collector. Several other types of package are available for low-power and high-power devices.

IC Packages

Like semiconductor devices, integrated circuits must be packaged to provide mechanical protection and terminals for electrical connection. Several standard packages in general use are illustrated in Fig. 7-28.

The metal-can-type of container affords electromagnetic shielding for the IC chip which cannot be obtained with the plastic packages (Fig. 7-28a). Note that the terminals are numbered clockwise from the tab, looking at the bottom of the can.

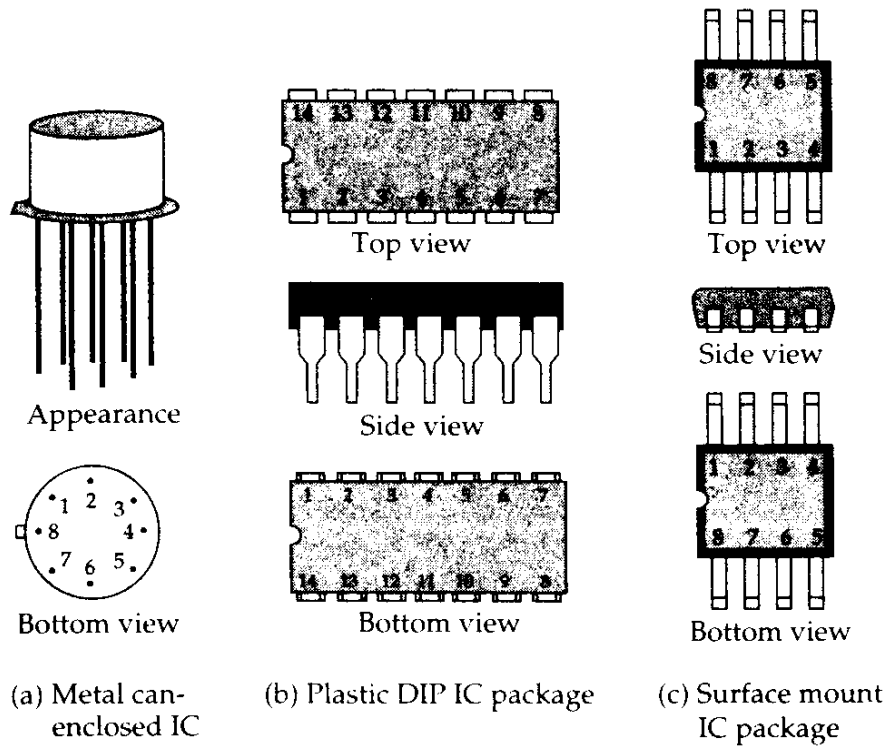


Figure 7-28 Integrated circuits are usually packaged in metal cans, dual-in-line plastic (DIP), or surface-mount plastic enclosures.

Plastic *dual-in-line* (DIP) packages (Fig. 7-28b) are more economical than metal cans and are widely used for industrial and consumer applications. The terminal numbering system again follows the bottom-view clockwise rule, starting from a notch in the plastic. A dimple, or other marker, is sometimes used instead of a notch. This is located on top of the package close to pin 1. DIP-type packages are much more convenient for circuit board use than cans, because the lead arrangement and flat package allow greater circuit densities. The surface mounting package shown in Fig. 7-28c is most economical for printed circuit boards because the terminals are soldered directly onto the board without the need for drilling holes.

Section 7-7 Review

- 7-7.1 Sketch the terminal arrangements for TO-92, TO-5, and TO-3 transistor packages, and identify the emitter, base, and collector terminals on each package.
- 7-7.2 Sketch metal can and DIP integrated circuit packages, and show the terminal numbering systems.

Review Questions

Section 7-1

- 7-1 Describe the process of preparing semiconductor material for device manufacture.

- 7-2** Explain diffusion and epitaxial growth, and discuss the application of each process to transistor manufacture.

Section 7-2

- 7-3** Draw sketches to show the construction of alloy diodes and diffused diodes. Briefly explain in each case.

Section 7-3

- 7-4** Explain the various transistor manufacturing requirements for maximum performance with respect to (a) current gain, (b) power dissipation, (c) frequency response, (d) switching response, and (e) breakdown voltage.

Section 7-4

- 7-5** Describe the microalloy and microalloy-diffusion techniques for transistor manufacture. Explain the advantages and disadvantages of transistors made by these techniques.
- 7-6** Using sketches, explain diffused mesa and epitaxial mesa transistors. Discuss the advantages and disadvantages of mesa transistors.
- 7-7** Explain the manufacturing process for diffused-planar and annular transistors, and discuss the advantages and disadvantages of each.
- 7-8** Show that most of the power dissipation in a transistor occurs at the collector-base junction.

Section 7-5

- 7-9** Briefly explain the thin-film and thick-film methods of integrated circuit manufacture, and discuss their advantages and disadvantages.
- 7-10** Using illustrations, explain the fabrication process for monolithic integrated circuits. Discuss the advantages and disadvantages of monolithic ICs.
- 7-11** Describe hybrid integrated circuits.

Section 7-6

- 7-12** Sketch the construction and circuit diagram for two monolithic integrated circuit transistors showing the parasitic components. State any precautions necessary in the use of the circuit.
- 7-13** Briefly explain how diodes, resistors, and capacitors are fabricated in monolithic integrated circuits.
- 7-14** Sketch the circuit of a current mirror, and explain how it functions.

Section 7-7

- 7-15** Sketch TO-92, TO-18, and TO-3 transistor packages, and identify the emitter, base, and collector terminals.
- 7-16** Sketch metal-can and DIP integrated circuit packages, and show the terminal numbering system.

CHAPTER 8

BJT Specifications and Performance

Objectives

You will be able to:

- 1 Identify important parameters on transistor data sheets for small-signal, high-power, high-frequency, and switching transistors, and determine the parameter values.
- 2 Calculate circuit power gains and power output changes in decibels.
- 3 Define Miller effect. Using the data sheet specifications, determine the input capacitance for transistor circuits.
- 4 Explain the effects of input and stray capacitances on BJT circuit frequency response, and calculate cutoff frequencies for CE, CB, and CC circuits.
- 5 Determine transistor switching times from their data sheets, and calculate device turn-on and turn-off times.
- 6 Define noise figure and noise factor, and estimate circuit output noise levels originating from resistors and transistors.
- 7 Using power derating factors and derating graphs, determine transistor maximum power dissipations at any temperature.
- 8 Sketch a thermal resistance circuit for a transistor with a heat sink. Determine the required thermal resistance for transistor heat sinks, and select suitable heat sinks for given circuit conditions.

INTRODUCTION

The performance characteristics and parameters for each type of transistor are specified on a data sheet published by the device manufacturer. The specifications must be correctly interpreted to achieve optimum transistor performance and to avoid breakdowns. Transistors must be selected to have a specified maximum collector-emitter voltage greater than the voltage applied to the CE terminals. The specified maximum collector current must be greater than the highest I_C level that may flow in the circuit. The maximum power that may be dissipated in a device is normally listed for a case temperature of

25°C. This value must be derated for operation at higher temperatures, and heat sinks must be employed where appropriate. Transistor cutoff frequency, switching times, noise figure, and input capacitance can all be very important in various applications.

8-1 TRANSISTOR DATA SHEETS

When selecting a transistor for a particular application, the data sheets provided by device manufacturers must be consulted. Typical transistor data sheets are shown in Appendices A-5 to A-9, and portions are reproduced in this chapter.

Most data sheets start with the device type number at the top of the page, a descriptive title, and a list of major applications for the device. This information is usually followed by mechanical data in the form of an illustration showing the appearance of the package, as well as indicating which leads are collector, base, and emitter (see Fig. 8-1).

The absolute maximum ratings of the transistor at a temperature of 25°C are listed next. These are the maximum voltages, currents, and power dissipation

 2N3903, 2N3904 (Silicon) NPN Transistors NPN silicon annular transistors, designed for general purpose switching and amplifier applications.			
E B C Maximum ratings at $T_A = 25^\circ\text{C}$ unless otherwise noted.			
Characteristic	Symbol	Rating	Unit
Collector-emitter voltage	V_{CEO}	40	V_{dc}
Emitter-base voltage	V_{EB}	6	V_{dc}
Collector current	I_C	200	mA

Figure 8-1 Portion of 2N3903 and 2N3904 transistor data sheet listing maximum terminal voltages and collector current.

that the device can survive without breaking down. It is very important that these ratings never be exceeded; otherwise, failure of the device is quite possible. *For reliability, the maximum ratings should not even be approached.* The maximum transistor ratings must also be adjusted downward for operation at temperatures greater than 25°C.

After the absolute maximum ratings, the data sheet normally shows a complete list of electrical characteristics for the device. (See data sheet A-5 in Appendix A.) Again, these are specified at 25°C, and allowances are necessary for temperature variations.

A complete understanding of all the quantities specified on a data sheet will not be achieved until circuit analysis and design are studied. Some of the most important quantities are considered below. It is important to note that the ratings for transistors are stated for specified circuit conditions. The ratings are no longer valid if these conditions change.

V_{CBO}	Collector-base voltage—maximum dc voltage for reverse-biased collector-base junction.
V_{CEO}	Collector-emitter voltage—maximum collector-emitter dc voltage with base open-circuited.
V_{EB0}	Emitter-base voltage—maximum emitter-base reverse-bias dc voltage.
I_C	Collector current—maximum dc collector current.
I_{CBO} or I_{CO}	Collector cutoff current—dc collector current with the collector-base junction reverse biased and the emitter open-circuited.
$V_{CE(sat)}$	Collector-emitter saturation voltage—collector-emitter voltage with the device in saturation.
h_{FE}	Static forward current transfer ratio—common emitter ratio of dc collector current and base current.
NF	Noise figure—ratio of total noise output to total noise input expressed as a decibel (dB) ratio. Defines the amount of noise added by the device.
f_{hfe} or f_{ae}	Common-emitter cutoff frequency—common-emitter operating frequency at which the device current gain falls to 0.707 of its normal (mid-frequency) value.
f_{hfb} or f_{ab}	Common-base cutoff frequency—as above, for common base.

From the section of the 2N3903 and 2N3904 data sheet reproduced in Fig. 8-1, the maximum collector-emitter voltage is specified as $V_{CEO} = 40$ V. Obviously, this device should not be used in any circuit with a supply exceeding 40 V; otherwise the transistor might break down. Preferably, the circuit supply voltage should always be less than the specified maximum V_{CEO} .

The maximum V_{EB} is listed as 6 V. This is the highest voltage that may be applied in reverse across the base-emitter junction of the transistor. If a greater voltage is used, the base-emitter junction of the transistor is likely to break down. The maximum (reverse) V_{EB} for most transistors is 5 V. In circumstances where the applied voltage might exceed this level, a clipping circuit may be used (see Sections 3-8 and 3-9). Alternatively, a diode can be connected in

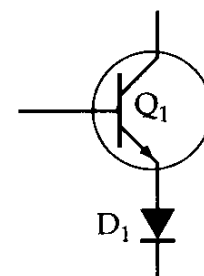


Figure 8-2 A diode connected in series with a transistor emitter protects the BE junction against excessive reverse voltage.

series with the transistor emitter to increase the breakdown voltage, as shown in Fig. 8-2. The normal forward-biased base-emitter voltage for a transistor is, of course, approximately 0.7 V for a silicon transistor and approximately 0.3 V for a germanium transistor.

Figure 8-3 shows sections of a transistor data sheet which specify the *dc current gain*, h_{FE} , and the *small signal* (or *ac*) *current gain*, h_{fe} . For a 2N3904 device with $I_C = 10$ mA, h_{FE} is listed as 100 minimum and 300 maximum (see the solid underlined quantities). If the transistor collector current is to be about 1 mA, then the minimum h_{FE} is 70 (dashed underline). The maximum h_{FE} for $I_C = 1$ mA is not specified. It is unlikely that $h_{FE(max)}$ would exceed 300 for this condition. However, because the manufacturer does not list a maximum value, it remains uncertain. This situation illustrates the importance of reliable bias circuits that make I_C and V_{CE} largely independent of h_{FE} (see Chapter 5).

The *small-signal current gain* (h_{fe}) for a 2N3904 is listed in Fig. 8-3 as 100 minimum and 400 maximum. Once again, this parameter is specified for a particular set of bias conditions. The signal frequency at which h_{fe} was measured is specified as 1 kHz. Section 8-3 discusses the fact that h_{fe} decreases at high frequencies.

2N3903, 2N3904					
Characteristic		Symbol	Min	Max	Unit
DC current gain		h_{FE}			—
$(I_C = 0.1$ mA dc, $V_{CE} = 1$ V dc)	2N3903		20	—	
	2N3904		40	—	
$(I_C = 1.0$ mA dc, $V_{CE} = 1$ V dc)	2N3903		35	—	
	2N3904		70	—	
$(I_C = 10$ mA dc, $V_{CE} = 1$ V dc)	2N3903		50	150	
	2N3904		100	300	
$(I_C = 50$ mA dc, $V_{CE} = 1$ V dc)	2N3903		30	—	
	2N3904		60	—	
$(I_C = 100$ mA dc, $V_{CE} = 1$ V dc)	2N3903		15	—	
	2N3904		30	—	
Small signal current gain		h_{fe}			—
$(I_C = 1$ mA dc, $V_{CE} = 10$ V, $f = 1$ kHz)					
	2N3903		50	200	
	2N3904		100	400	

Figure 8-3 Portion of 2N3903 and 2N3904 transistor data sheet listing dc current gain and small-signal ac current gain.

Example 8-1

Referring to the 2N3905 specification in Appendix A-6, determine the h_{FE} range for $I_C = 10$ mA, and the h_{fe} range for $f = 1$ kHz. Also determine the emitter-base breakdown voltage.

Solution

$$\text{At } I_C = 10 \text{ mA,} \quad h_{FE} = 50 \text{ to } 150$$

$$\text{At } f = 1 \text{ kHz,} \quad h_{fe} = 50 \text{ to } 200$$

$$V_{(BR)EBO} = 5 \text{ V}$$

Practice Problem

8-1.1 From the data sheet A-7 in Appendix A, determine the following quantities for a 2N3251 transistor: $h_{FE(\min)}$ at $I_C = 1$ mA, $h_{fe(\min)}$ at $f = 1$ kHz, $V_{CE(\max)}$, $V_{EB(\max)}$, and noise figure.

8-2 DECIBELS AND HALF-POWER POINTS**Power Measurement in Decibels**

The *power gain* (A_p) of an amplifier may be expressed in terms of the log of the ratio of output power (p_o) to input power (p_i). This is illustrated in Fig. 8-4.

$$A_p = \log_{10} \left(\frac{p_o}{p_i} \right) \text{ bels}$$

or

$$A_p = 10 \log_{10} \left(\frac{p_o}{p_i} \right) \text{ decibels (dB)} \quad (8-1)$$

Thus, the *decibel* is the unit of *power gain*, or *power level change*.

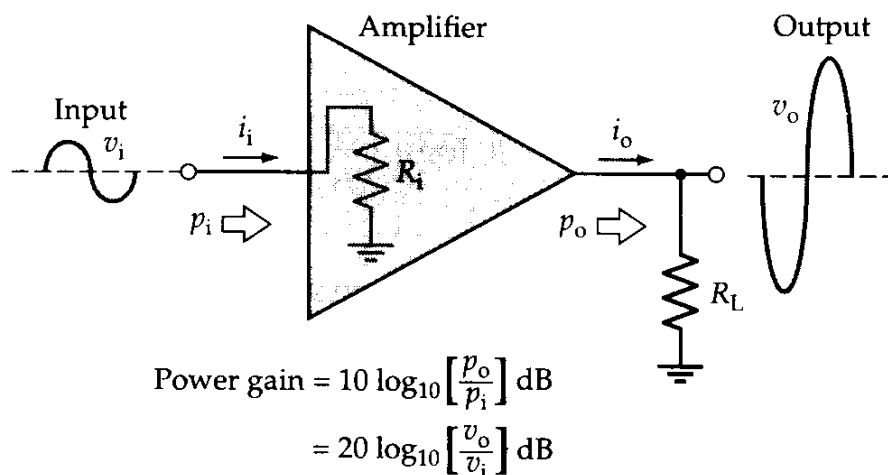


Figure 8-4 Amplifier power gain is measured in decibels (dB). Output power level changes are also measured in decibels.

The power dissipated in a resistance is (v^2/R) .

$$\text{So} \quad A_p = 10 \log_{10} \left(\frac{(v_o)^2/R_L}{(v_i)^2/R_i} \right) \text{ dB}$$

$$\text{When } R_i = R_L, \quad A_p = 10 \log_{10} \left(\frac{v_o}{v_i} \right)^2 \text{ dB}$$

$$\text{or} \quad A_p = 20 \log_{10} \left(\frac{v_o}{v_i} \right) \text{ dB} \quad (8-2)$$

Also, $P_o = i_o^2 R_L$ and $P_i = i_i^2 R_i$

$$\text{So} \quad A_p = 20 \log_{10} \left(\frac{i_o}{i_i} \right) \text{ dB} \quad (8-3)$$

It is important to note that *the above equations are correct only when the load resistance connected to the circuit output is equal to the circuit input resistance*. However, in the electronics industry, these equations are sometimes applied regardless of the relationship between input and load resistances!

Amplifier output power level changes (ΔP_o) are also measured in decibels, and in this case the equations become

$$\Delta P_o = 10 \log_{10} \left(\frac{p_2}{p_1} \right) \text{ dB} \quad (8-4)$$

where p_1 is the initial output power level and p_2 is the new level.

$$\text{Also,} \quad \Delta P_o = 20 \log_{10} \left(\frac{v_2}{v_1} \right) \text{ dB} \quad (8-5)$$

$$\text{and} \quad \Delta P_o = 20 \log_{10} \left(\frac{i_2}{i_1} \right) \text{ dB} \quad (8-6)$$

where v_1 and i_1 are the initial output voltage and current levels, and v_2 and i_2 are the new levels. Equations 8-4 to 8-6 are correct so long as the amplifier load resistance remains constant. Examples 8-2 and 8-3 demonstrate that the output power of an amplifier is reduced by 3 dB when the measured power falls to half its normal level or when the measured voltage falls to 0.707 of its normal level.

Example 8-2

The output power from an amplifier is 50 mW when the signal frequency is 5 kHz. The power output falls to 25 mW when the frequency is increased to 20 kHz. Calculate the output power change in decibels.

Solution

$$\begin{aligned}\text{Eq. 8-4: } \Delta P_o &= 10 \log_{10} \left(\frac{p_2}{p_1} \right) = 10 \log_{10} \left(\frac{25 \text{ mW}}{50 \text{ mW}} \right) \\ &= -3 \text{ dB}\end{aligned}$$

Half-Power Points

Figure 8-5 shows a typical graph of amplifier output voltage or power plotted versus frequency. Note that the frequency scale is logarithmic. The output normally remains constant over a middle range of frequencies and falls off at low and high frequencies owing to effects explained in Sections 8-3 and 8-4. The gain over this middle range is termed the *mid-frequency gain*. The low frequency and high frequency at which the gain falls by 3 dB are designated f_1 and f_2 , respectively. This range (from f_1 to f_2) is normally considered the useful range of operating frequency for the amplifier, and the frequency difference ($f_2 - f_1$) is termed the *amplifier bandwidth (BW)*.

Frequencies f_1 and f_2 are termed the *half-power points*, or the *3 dB points*. This is because, as shown in Ex. 8-2, the power output is -3 dB from its normal level when p_2 is half p_1 . When the amplifier output is expressed as a voltage on the graph of frequency response, the 3 dB points (f_1 and f_2) occur when v_2 is $0.707v_1$. This relationship is demonstrated in Ex. 8-3.

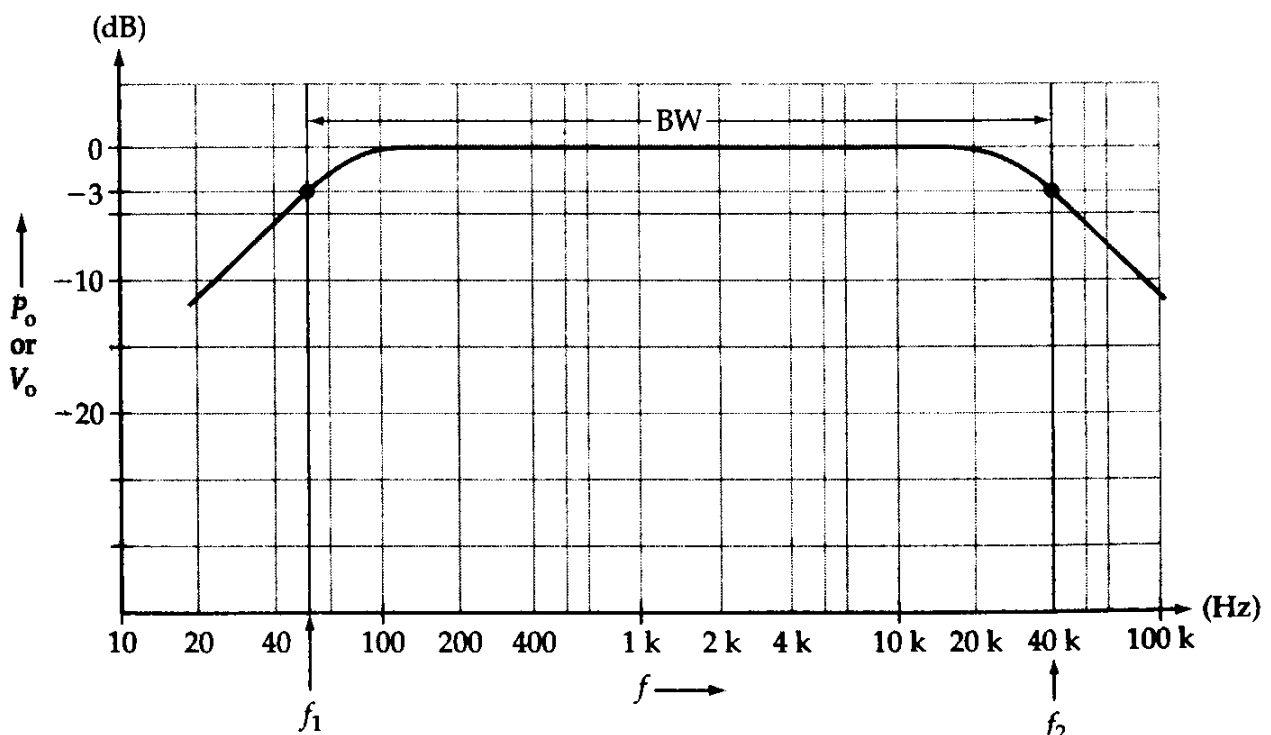


Figure 8-5 The output voltage and output power of an amplifier normally remain constant over a middle band of signal frequencies and fall off at high and low frequencies.

Example 8-3

The output voltage of an amplifier is measured as 1 V at 5 kHz and as 0.707 V at 20 kHz. Calculate the output power change.

Solution

$$\begin{aligned}\text{Eq. 8-5: } \Delta P_o &= 20 \log_{10} \left(\frac{v_2}{v_1} \right) = 20 \log_{10} \left(\frac{0.707 \text{ V}}{1 \text{ V}} \right) \\ &= -3 \text{ dB}\end{aligned}$$

Practice Problem

8-2.1 The output voltage of an amplifier is 2 V when the signal frequency is 3 kHz, and 0.5 V when the signal frequency is increased to 50 kHz. Calculate the decibel change in output power.